

Healthcare IT Fundamentals – Section 2,

Lecture 1

Electronic Health Records (EHR) & Clinical Workflow (Simple Notes)

Big Picture

In Section 1, we learned:

- Why healthcare became digital
- HIPAA and compliance
- EHR, EMR, FHIR, HL7, ICD-10, etc.

Now we learn:

How an EHR actually works inside a hospital.

Think of an EHR not as a database, but as a complete operating system for a hospital.

It manages everything from:

```
graph TD; A[Patient Registration] --> B[Doctor Consultation]; B --> C[Lab Tests]; C --> D[Medication]; D --> E[Billing]; E --> F[Discharge];
```

1. What is an Enterprise EHR?

Many people think an EHR is one application.

Reality:

An EHR is a collection of many integrated modules.

Think of it like:

Rails Monolith with Many Engines

or

A Microservice Ecosystem

Each module handles a different responsibility.

Main Components of an EHR

Clinical Documentation

Where doctors and nurses record:

- Patient history
 - Progress notes
 - Symptoms
 - Diagnoses
 - Vital signs
-

Old World

Paper chart

New World

Results Management

Stores and displays:

- Lab results
 - Blood reports
 - X-rays
 - MRI reports
 - Radiology reports
-

Example

Doctor orders:

CBC Blood Test

Lab completes it.

Result automatically appears in EHR.

Order Management

Handles requests for:

- Medications
 - Lab tests
 - Imaging studies
-

Example:

Doctor orders:

Chest X-Ray

System routes request to radiology department.

Decision Support Engine

The "smart brain" inside EHR.

Checks for:

- Drug interactions
 - Allergies
 - Wrong dosage
 - Missing screenings
-

Example

Doctor prescribes:

Penicillin

Patient has allergy.

System immediately warns:

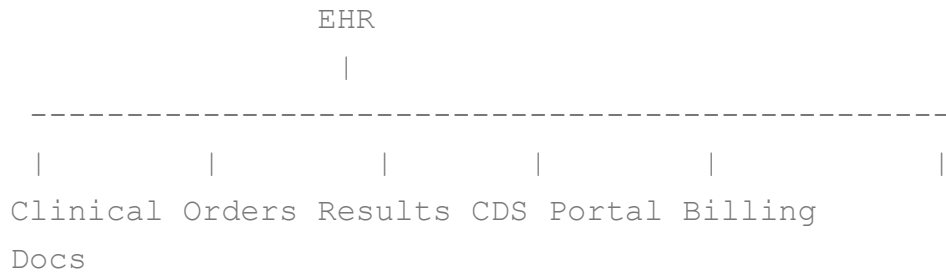
ALLERGY ALERT

Patient Support

Patient-facing tools:

- Patient portals
 - Health education
 - Appointment scheduling
 - Secure messaging
-

EHR Architecture Overview



All modules work together.

2. CPOE

Computerized Provider Order Entry

One of the most important EHR features.

Before CPOE

Doctor writes:

Take Drug X

Handwriting unclear.

Pharmacist misreads it.

Result:

- ✗ Medication errors
 - ✗ Wrong dosage
 - ✗ Delayed treatment
-

After CPOE

Doctor enters order electronically.

Drug X
500 mg
Twice daily

Order instantly reaches:

- Pharmacy
 - Lab
 - Radiology
-

Benefits of CPOE

Better Safety

Checks:

- Allergies
 - Drug interactions
 - Dosage issues
-

Faster Processing

No paper transport.

Order reaches destination immediately.

Better Tracking

System knows:

- Who ordered
 - When ordered
 - Current status
-

Software Engineer Analogy

Before:

Paper request

After:

API Request

CPOE works like sending structured requests through APIs.

3. Patient Portals

Patient Portal = Patient's self-service website.

Patients Can

View:

- Lab results
 - Medications
 - Visit summaries
 - Diagnoses
-

Schedule:

- Appointments
-

Pay:

- Bills
-

Communicate:

- Secure messages
-

Example

Instead of calling hospital:

`What was my cholesterol result?`

Patient simply logs in.

Benefits

Better Patient Engagement

Patients become active participants.

Reduced Call Volume

Fewer phone calls to staff.

Improved Convenience

Access records anytime.

Challenge

Portal must be easy to use.

Otherwise:

- Elderly patients
- Non-technical users

may struggle.

4. Clinical Decision Support (CDS)

One of the most important patient safety features.

What is CDS?

Software that helps doctors make better decisions.

Active CDS

Shows alerts.

Example:

```
Drug Interaction Warning
```

or

```
Patient Allergy Alert
```

Passive CDS

Shows recommendations.

Example:

```
Patient due for flu vaccine
```

or

```
Patient needs mammogram screening
```

Advanced CDS

Provides:

Order Sets

Predefined treatment packages.

Example

Patient diagnosed with:

Pneumonia

System suggests:

- Antibiotics
- Blood tests
- Chest X-ray

based on clinical guidelines.

Alert Fatigue

Too many alerts can be dangerous.

Example:

Alert
Alert
Alert
Alert
Alert

Doctors start ignoring them.

Eventually:

A critical alert may be missed.

This is called:

Alert Fatigue

5. Lab & Imaging Integration

Modern EHR integrates with:

- Laboratory systems
 - Radiology systems
 - Imaging systems
-

Bidirectional Interface

Means:

Send Data

Doctor sends order.

Receive Data

Lab sends result back.

Example:

```
Doctor → Lab  
      ← Result
```

Two-way communication.

Benefits

No manual entry.

No lost reports.

No duplicate testing.

Example

Without integration:

`Where is the blood report?`

Nobody knows.

Test repeated.

With integration:

Result appears automatically.

6. PACS

Picture Archiving and Communication System

Stores medical images.

Examples:

- X-ray
 - MRI
 - CT Scan
 - Ultrasound
-

Before

Images stored physically.

Hard to access.

Now

Doctors can instantly open images from EHR.

Think:

Google Photos

for medical images.

7. Trending Data

Because data is digital:

System can show trends.

Example:

Diabetes patient.

Glucose readings:

Jan → 150

Feb → 145

Mar → 130

Apr → 120

Doctor sees improvement.

This is difficult with paper charts.

8. Administrative Modules

Not directly related to treatment.

But essential for hospital operations.

Registration

Patient enters hospital.

Staff collects:

- Name
 - Insurance
 - Contact details
-

Creates patient record.

ADT

Admission, Discharge, Transfer

Tracks patient status.

Admission

Patient enters hospital.

Transfer

Patient moves departments.

Example:

ICU → General Ward

Discharge

Patient leaves hospital.

ADT acts as:

Source of truth for patient location.

Scheduling

Manages:

- Doctor appointments
 - Operating rooms
 - MRI machines
 - Hospital resources
-

Example:

System ensures:

MRI machine

is not booked for two patients simultaneously.

Billing & Coding

Links clinical care to reimbursement.

Doctor records:

Diabetes

System maps it to:

ICD-10 code.

Insurance claim generated.

Benefits

Faster reimbursement.

Fewer billing errors.

Reporting Dashboards

Used by hospital leadership.

Shows:

- Patient volume
 - Revenue
 - Readmission rates
 - Wait times
 - Quality metrics
-

Helps management make decisions.

Patient Journey Through EHR

Registration



ADT Creation



Doctor Documentation



CPOE Orders



Lab / Radiology



Results Returned



CDS Checks



Treatment

↓
Billing & Coding
↓
Discharge
↓
Patient Portal Access

Quick Revision Sheet

Clinical Documentation

Stores notes, history, diagnoses.

Results Management

Displays lab and imaging results.

Order Management

Handles medication and test requests.

CPOE

Electronic ordering system.

Patient Portal

Patient access to records and communication.

CDS

Clinical Decision Support.

Provides alerts and recommendations.

Alert Fatigue

Too many alerts causing users to ignore them.

PACS

Stores and displays medical images.

ADT

Admission, Discharge, Transfer.

Scheduling

Manages appointments and resources.

Billing & Coding

Converts care into insurance claims.

For a Ruby on Rails Developer

Think of an EHR as:

Software Concept	Healthcare Equivalent
User Dashboard	Patient Portal
Admin Panel	Hospital Operations
Service Layer	CDS Engine
API Requests	CPOE Orders

Software Concept	Healthcare Equivalent
File Storage	PACS Images
Analytics Dashboard	Reporting Module
Authentication	HIPAA Access Control
Workflow Engine	Clinical Workflow

One-line takeaway:

An EHR is not just a database—it is a workflow platform that manages patient care, orders, results, billing, and communication from registration to discharge. 🚀